VAYUBUDHI ENVIRONMENTAL INTELLIGENCE SYSTEM

Machine Learning & Real-Time Telemetry Technical Dossier

Comprehensive Architectural Specification, Physics Derivations, and Model Evaluation Benchmarks

1. Platform Architecture Overview

VayuBudhi is an enterprise environmental AI platform that fuses real-time multi-source telemetry, continuous atmospheric conservation physics, gradient-boosted temporal forecasting, and physics-informed source apportionment.

```
+------------------------------------------------------------------------------+ | DATA INGESTION LAYER | | [Satellite CAMS Model] [TomTom Live Traffic] [ESP32 Hardware Edge Lab] | | (PM2.5, NO2, SO2, CO, O3) (Arterial Congestion) (SPS30, SGP41, SCD41, BME280) | +-------------------------+-------------------------+------------------------------+ | | v v +------------------------------------------------------------------------------+ | ATMOSPHERIC PHYSICS & CONSERVATION ENGINE | | * Barometric Hydrostatic Altitudinal Scaling: (1013.25 / Pressure)^2.70 | | * Planetary Boundary Layer (PBLH) Box-Model Convective Ventilation | | * Secondary Organic Aerosol (SOA) Photochemical Formation | | * 2D Inverse Distance Squared Weighting (IDW, p=2) & Circular Wind Vectors | +-----------------------------------------+--------------------------------------+ | v +------------------------------------------------------------------------------+ | MACHINE LEARNING & UNCERTAINTY CALIBRATION | | [24h / 48h / 72h Forecasters] [Source Apportionment & PINN] | | - City-Specific XGBoost Regressors - Random Forest & CatBoost Classifier | | - 90% Certified MAPIE Conformal Sets - 90% Conformal Prediction Sets | | - Online Kalman/EMA Bias Tracking - Gaussian Plume Upwind Triangulation | +-----------------------------------------+--------------------------------------+ | v +------------------------------------------------------------------------------+ | FASTAPI PRODUCTION SERVING LAYER | | /api/city-data | /api/live | /api/attribution | /api/forecast | /api/report | +------------------------------------------------------------------------------+
```

2. Machine Learning Forecasting Models

2.1 Model Pipeline & Feature Representation

The platform deploys multi-horizon **Gradient Boosted Decision Trees (XGBoost, LightGBM, CatBoost)** to predict ambient PM2.5 concentrations at 24-hour, 48-hour, and 72-hour intervals. Every model takes an identical 7-dimensional physical feature vector:

```
Input Vector = [ PM2.5, PM10, Temperature (°C), Relative Humidity (%), Barometric Pressure (hPa), Wind Speed (m/s), Boundary Layer Height (m) ]
```

CITY-SPECIFIC TAILORED MODELS

Trained specifically on historical meteorological and ambient sensor distributions for major cities (**Delhi, Hyderabad, Bengaluru**). Captures local microclimate features like Delhi's winter thermal inversions and Bengaluru's high-altitude convective mixing.

UNIFIED NATIONAL BASE MODELS

Trained on aggregated multi-city historical records (over 650,000 telemetry rows). Acts as a high-robustness general model across secondary cities (Kolkata, Mumbai, Pune, Chennai, Ahmedabad, Jaipur, Lucknow, Chandigarh).

2.2 Accuracy Benchmarks & Evaluation Metrics

Models were evaluated on strictly chronologically held-out test datasets (70% train, 15% conformal calibration, 15% test) with zero data leakage:

FORECAST HORIZON	MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARED ERROR (RMSE)	DETERMINATION COEFF. (R^2)	PERSISTENCE BASELINE RMSE	ERROR REDUCTION VS. BASELINE	90% CERTIFIED CONFORMAL COVERAGE
24-Hour Ahead	**6.90 μg/m³**	9.41 μg/m³	0.472	13.03 μg/m³	**+27.7% Improvement**	**94.0%**
48-Hour Ahead	**8.15 μg/m³**	11.05 μg/m³	0.274	14.10 μg/m³	**+21.7% Improvement**	**92.7%**
72-Hour Ahead	**8.63 μg/m³**	11.49 μg/m³	0.215	14.10 μg/m³	**+18.5% Improvement**	**93.9%**

METRIC INTERPRETATIONS

- **MAE = 6.90 μg/m³**: On average, the 24-hour prediction deviates by less than 7 μg/m³ from actual next-day sensor truth.
- **Persistence Baseline Comparison**: The standard meteorological assumption assumes tomorrow's air quality will equal today's. VayuBudhi outperforms this benchmark by **27.7%** at 24 hours.

3. Uncertainty Quantification & Online Adaptation

3.1 Certified Uncertainty: MAPIE Split Conformal Prediction

Standard neural networks and regression models produce point estimates without quantifying certainty. VayuBudhi wraps every forecaster and classifier with **MAPIE (Model Agnostic Prediction Interval Estimator)** based on inductive conformal prediction theory:

1. **Non-Conformity Residuals**: On a separate calibration dataset, compute residual: `R_i = |y_i - y_hat(x_i)|`
2. **Conformal Quantile**: For a 90% confidence target ($\alpha = 0.10$), extract the empirical $(1 - \alpha)$ quantile `q_hat` of residuals.
3. **Guaranteed Interval**: Generates dynamic interval: `[ y_hat(x_new) - q_hat, y_hat(x_new) + q_hat ]`

Guaranteed Result: The probability that the true future PM2.5 concentration falls inside the generated interval is provably ≥ 90% (empirically achieves **94.0% coverage**).

3.2 Closed-Loop Kalman / EMA Online Bias Adaptation

In live production serving, ambient weather conditions can shift unseasonally. The backend maintains an active verification log:

- As past predictions mature each hour, the backend calculates: `Residual = Actual_PM25 - Predicted_PM25`
- Updates a recursive Exponential Moving Average (EMA) bias tracker ($\alpha = 0.35$):
  `Bias_t = 0.35 * Residual_t + 0.65 * Bias_{t-1}`
- Subsequent live forecasts are dynamically adjusted by this bias, eliminating systematic drift.

4. Source Apportionment, Explainability & PINN Plume Inversion

4.1 Multi-Class Source Classifier

The source attribution model classifies ambient pollution into 4 physical emission sectors:

- **Vehicular Exhaust & Transport**: High NO_2, traffic congestion slowdowns, fine-to-coarse ratio PM2.5 / PM10 > 0.65.
- **Industrial Point Sources**: High SO_2, elevated PM10, proximity to industrial zoning.
- **Biomass & Crop Residue Combustion**: Fine fraction PM2.5 / PM10 > 0.80, low temperature, elevated CO.
- **Road Dust Resuspension**: Low relative humidity < 40%, coarse particle fraction dominant PM2.5 / PM10 < 0.45.

4.2 TreeSHAP Feature Explainability

Using cooperative game theory (Shapley values), the system explains exactly *why* pollution is rising or falling for any selected ward:

```
• Particulate Mass (PM2.5): +18.4 µg/m³ (Primary mass loading driver)
• Thermal Inversion Entrapment: +6.8 µg/m³ (Shallow 420m PBLH ceiling traps ground soot)
• SGP41 VOC Gaseous Precursor: +3.2 µg/m³ (Hydrocarbon exhaust driving secondary nucleation)
• Horizontal Wind Ventilation: -4.5 µg/m³ (2.8 m/s wind mitigating localized accumulation)
```

4.3 Physics-Informed Neural Network (PINN) Plume Inversion

When a localized sensor spike is detected, the physics engine inverts the classical **Gaussian Plume atmospheric advection-diffusion equation**:

```
C(x, y, z) = [ Q / (2 π u σ_y σ_z) ] * exp( -y² / (2 σ_y²) ) * [ exp( -(z - H)² / (2 σ_z²) ) + exp( -(z + H)² / (2 σ_z²) ) ]
```

Upwind Source Triangulation Algorithm:

1. Extracts meteorological wind direction θ (direction from which the wind arrives).
2. Calculates the 15-minute upwind travel vector: `Distance = min(Wind_Speed * 900s, 1500m)`.
3. Projects backward coordinates on Earth's surface: `Delta_Lat = (Distance * cos(θ)) / 111,000m` and `Delta_Lon = (Distance * sin(θ)) / (111,000m * cos(Lat))`.
4. Pinpoints the estimated source coordinates (Lat, Lon) and computes emission rate `Q = C * (2 π u σ_y σ_z)` in grams per second.

5. Live Telemetry & Underground Physics Calculations

5.1 Ingested External APIs

EXTERNAL SERVICE	INGESTED PARAMETERS	UPDATE CADENCE & FUNCTION
Open-Meteo Air Quality (CAMS)	PM2.5, PM10, NO2, SO2, CO, O3, Aerosol Optical Depth (AOD), Desert Dust	Continuous hourly background chemical baseline
Open-Meteo Weather API	Planetary Boundary Layer Height (PBLH), Barometric Pressure, Temp, Humidity, Wind Vector	Real-time boundary layer physics and wind particle synthesis
TomTom Traffic Flow API	Current speed, Free-flow speed, Travel delay, Congestion score	Dynamic traffic slowdown index for street-canyon vehicle soot injection
NASA FIRMS Satellite	Thermal anomaly coordinates, Brightness temperature, Fire Radiative Power	Cross-verification of agricultural and open biomass combustion
OpenStreetMap (OSM)	Administrative district polygons, Industrial zoning corridors	High-resolution ward boundaries and land-use spatial masking
Hardware Lab (ESP32)	Sensirion SPS30 (PM1.0/2.5/4.0/10), SGP41 (VOC/NOx), SCD41 (CO2), BME280 (T/H/P)	Physical edge sensor calibration and live hardware testbench

5.2 Continuous Atmospheric Conservation Engine

Raw satellite models have coarse 10 km grid resolution. VayuBudhi's backend conservation engine transforms them into microclimate ground truth:

1. Barometric Hydrostatic Altitudinal Scaling

Barometric air density compresses aerosol volumetric mass in low-lying basins:

```
Altitudinal Factor = ( 1013.25 / max(700.0, Surface_Pressure) )^2.70
```

2. Planetary Boundary Layer Height (PBLH) Box Model

During morning inversions (08:00 - 10:30 IST), the ceiling contracts to 350m - 500m, trapping smoke. In the afternoon, thermal convection expands the ceiling to 2000m, flushing the basin:

```
Ventilation Factor = ( 800.0 / clamp(PBLH, 350, 3000) )^0.32 * ( 2.8 / max(0.8, Wind_Speed) )^0.22
```

3. Secondary Organic Aerosol (SOA) Photochemical Formation

NO2 and SO2 gases chemically react under sunlight to nucleate into fine particulate mass:

```
SOA Formation = ( NO2 * 0.15 + SO2 * 0.15 ) * min(1.2, Ventilation Factor)
```

4. Street-Canyon Dynamic Traffic Emission Injection

```
Congestion Index = max( 0.0, (FreeFlow_Speed - Current_Speed) / FreeFlow_Speed )
Traffic Injection = 14.0 * Congestion Index * Ventilation Factor
```

5. Final Calibrated Ground PM2.5

```
Final PM2.5 = max( 8.0, (Raw_PM25 * Altitudinal Factor * Ventilation Factor * 0.70) + Urban_Baseline +
```

```
SOA Formation + Traffic Injection )
```

5.3 Spatial Inverse Distance Weighting (IDW) & Circular Wind Vectors

To interpolate continuous pollution across any ward centroid (x0, y0) from surrounding monitoring stations:

```
Z_estimated = Sum( w_i * Z_i ) / Sum( w_i ), where w_i = 1 / ( Distance_i² + ε )
```

Circular Wind Direction Averaging: Because compass angles wrap around at 360° = 0°, simple arithmetic averages fail (e.g. average of 350° and 10° is 0°/North, not 180°/South). The engine computes the weighted trigonometric circular mean:

```
S = Sum( w_i * sin(θ_i) ), C = Sum( w_i * cos(θ_i) )
Mean Wind Direction = ( atan2(S, C) * 180 / π + 360 ) mod 360
```

5.4 EPA Standard Breakpoint Conversion

PM2.5 BREAKPOINT RANGE (MG/M^3)	EPA AQI RANGE	CATEGORY
0.0 - 12.0	0 - 50	Good
12.1 - 35.4	51 - 100	Moderate
35.5 - 55.4	101 - 150	Unhealthy for Sensitive Groups
55.5 - 150.4	151 - 200	Unhealthy
150.5 - 250.4	201 - 300	Very Unhealthy
250.5 - 500.4	301 - 500	Hazardous